Chapter 4. Determinants

One number that decides invertibility and measures volume

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4.1 Determinants of order 2

Motivation. The number $ad - bc$ decides invertibility, gives signed area, and is linear in each row, the property that generalises.

Definition (Determinant of order 2)

For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{M}_{2\times 2}(\mathbb{F})$, $\det(A) = |A| = ad - bc$.

Example 1

For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & 2 \\ 6 & 4 \end{smallmatrix}\right)$: $\det A = -2$, $\det B = 0$, $\det(A+B) = \det\left(\begin{smallmatrix} 4 & 4 \\ 9 & 8 \end{smallmatrix}\right) = -4$: det is not linear.

Theorem 4.1

$\det : \mathrm{M}_{2\times 2}(\mathbb{F}) \to \mathbb{F}$ is linear in each row when the other row is fixed: for $u, v, w \in \mathbb{F}^2$, $k \in \mathbb{F}$,

$$\det \begin{pmatrix} u + kv \\ w \end{pmatrix} = \det \begin{pmatrix} u \\ w \end{pmatrix} + k \det \begin{pmatrix} v \\ w \end{pmatrix}, \qquad \det \begin{pmatrix} w \\ u + kv \end{pmatrix} = \det \begin{pmatrix} w \\ u \end{pmatrix} + k \det \begin{pmatrix} w \\ v \end{pmatrix}.$$

Proof. With $u = (a_1, a_2)$, $v = (b_1, b_2)$, $w = (c_1, c_2)$: $(a_1 + kb_1)c_2 - (a_2 + kb_2)c_1 = (a_1c_2 - a_2c_1) + k(b_1c_2 - b_2c_1)$. The second identity is analogous. ∎

Theorem 4.2

$A \in \mathrm{M}_{2\times 2}(\mathbb{F})$ is invertible iff $\det A \neq 0$, and then $A^{-1} = \frac{1}{\det A}\left(\begin{smallmatrix} A_{22} & -A_{12} \\ -A_{21} & A_{11} \end{smallmatrix}\right)$.

Proof. If $\det A \neq 0$, the displayed M satisfies $AM = MA = I$. Conversely, invertible A has $\operatorname{rank} A = 2$, so $A_{11} \neq 0$ or $A_{21} \neq 0$. If $A_{11} \neq 0$, adding $-A_{21}/A_{11}$·row 1 to row 2 gives $\left(\begin{smallmatrix} A_{11} & A_{12} \\ 0 & A_{22} - A_{12}A_{21}/A_{11} \end{smallmatrix}\right)$, still of rank 2 (Theorem 3.4), so $A_{22} - A_{12}A_{21}/A_{11} \neq 0$, i.e. $\det A \neq 0$. If $A_{21} \neq 0$, exchange the rows. ∎

4.1.1 The area of a parallelogram

For an ordered basis $\beta = \{u, v\}$ of $\mathbb{R}^2$, the **orientation** is $O\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) = \det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right) / \left|\det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)\right| = \pm 1$: $+1$ if β is right-handed (u turns counterclockwise by less than π to v), -1 if left-handed; $O = 1$ if $\{u, v\}$ is dependent. The **area** of the parallelogram with sides u, v is

$$\operatorname{area} \begin{pmatrix} u \\ v \end{pmatrix} = O \begin{pmatrix} u \\ v \end{pmatrix} \cdot \det \begin{pmatrix} u \\ v \end{pmatrix} = \left| \det \begin{pmatrix} u \\ v \end{pmatrix} \right|.$$

Reason. $\delta(u, v) = O \cdot \text{area}$ is linear in each row (shears preserve it, scaling a side scales it), vanishes on equal rows, and $\delta(e_1, e_2) = 1$, so $\delta = \det$ (Theorem 4.12, $n = 2$).

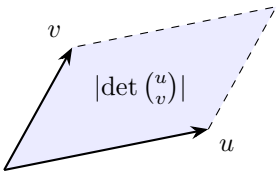


Figure. *Area* $|\det\left(\begin{smallmatrix} u \\ v \end{smallmatrix}\right)|$*; the sign records orientation.*

Example 2

For $u = (-1, 5)$, $v = (4, -2)$: $\det\begin{pmatrix} -1 & 5 \\ 4 & -2 \end{pmatrix} = -18$: area 18, left-handed.

4.2 Determinants of order n

Motivation. Defined by first-row expansion, det is linear in each row and expandable along any row, which governs its behaviour under row operations.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$, $\tilde{A}_{ij}$ is A with row i and column j deleted.

Definition (Determinant)

For $n = 1$, $\det(A_{11}) = A_{11}$. For $n \geq 2$,

$$\det(A) = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}).$$

$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$ is the **cofactor** of A_{ij}, and the formula is the **cofactor expansion along the first row**. For $n = 2$ it gives $A_{11}A_{22} - A_{12}A_{21}$.

Example 1

$A = \begin{pmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{pmatrix}$: $\det A = 1 \cdot (30 - 8) - 3 \cdot (18 + 8) + (-3)(-12 - 20) = 22 - 78 + 96 = 40$.

Example 2

$B = \begin{pmatrix} 0 & 1 & 3 \\ -2 & -3 & -5 \\ 4 & -4 & 4 \end{pmatrix}$: $\det B = 0 - 1 \cdot (-8 + 20) + 3 \cdot (8 + 12) = 48$.

Example 3

$C = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & -3 \\ -2 & -3 & -5 & 2 \\ 4 & -4 & 4 & -6 \end{pmatrix}$: only C_{11}, C_{14} contribute, with $\tilde{C}_{11} = A$, $\tilde{C}_{14} = B$, so $\det C = 2 \cdot 40 + (-1)^5 \cdot 1 \cdot 48 = 32$.

Example 4

$\det I_n = 1 \cdot \det I_{n-1} = 1$ by induction.

Theorem 4.3

The determinant is linear in each row when the other rows are fixed: if rows $a_1, \ldots, a_n$ have $a_r = u + kv$, then

$$\det(a_1, \ldots, u + kv, \ldots, a_n) = \det(a_1, \ldots, u, \ldots, a_n) + k \det(a_1, \ldots, v, \ldots, a_n).$$

Proof. Induction on n; $n = 1$ is clear. Let A, B, C have row r equal to $u + kv$, u, v, other rows equal.

Step 1 ($r = 1$). $\tilde{A}_{1j} = \tilde{B}_{1j} = \tilde{C}_{1j}$, so $\det A = \sum_j (-1)^{1+j}(u_j + kv_j)\det \tilde{A}_{1j} = \det B + k \det C$.

Step 2 ($r > 1$). $A_{1j} = B_{1j} = C_{1j}$, and $\tilde{A}_{1j}, \tilde{B}_{1j}, \tilde{C}_{1j}$ differ only in the row from row r ($u + kv$, u, v minus entry j). By induction $\det \tilde{A}_{1j} = \det \tilde{B}_{1j} + k \det \tilde{C}_{1j}$; multiply by $(-1)^{1+j} A_{1j}$ and sum. ■

Corollary

A matrix with a zero row has determinant 0.

Proof. If row r is $0 = 0 + 1 \cdot 0$, Theorem 4.3 gives $\det A = \det A + \det A$. ■

Lemma

If $n \geq 2$ and row i of $B \in \mathrm{M}_{n\times n}(\mathbb{F})$ is e_k, then $\det B = (-1)^{i+k} \det(\tilde{B}_{ik})$.

Proof. Induction on n; $n = 2$ is direct. Let $n \geq 3$; $i = 1$ is the definition, so let $i > 1$. For $j \neq k$, let C_{ij} be $\tilde{B}_{1j}$ minus row $i - 1$ (from row i of B) and the column through k. Row $i - 1$ of $\tilde{B}_{1j}$ is 0 ($j = k$), e_{k-1} ($j < k$), e_k ($j > k$), in $\mathbb{F}^{n-1}$. By induction,

$$\det \tilde{B}_{1j} = \begin{cases} 0 & j = k \\ (-1)^{(i-1)+(k-1)} \det C_{ij} & j < k \\ (-1)^{(i-1)+k} \det C_{ij} & j > k. \end{cases}$$

Hence $\det B = \sum_{j<k} (-1)^{1+j} B_{1j} (-1)^{i+k} \det C_{ij} + \sum_{j>k} (-1)^{1+j} B_{1j} (-1)^{i+k+1} \det C_{ij}$. Expanding $\det \tilde{B}_{ik}$ along its first row, B_{1j} sits in column j ($j < k$) or $j - 1$ ($j > k$) with minor C_{ij}:

$$\det \tilde{B}_{ik} = \sum_{j<k} (-1)^{1+j} B_{1j} \det C_{ij} + \sum_{j>k} (-1)^{j} B_{1j} \det C_{ij}.$$

Multiply by $(-1)^{i+k}$ to get $\det B$. ■

Theorem 4.4

The determinant can be expanded along any row: for each i, $\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$.

Proof. For $i = 1$ this is the definition; let $i > 1$. Row i of A is $\sum_j A_{ij} e_j$; let B_j be A with row i replaced by e_j. By Theorem 4.3 and the Lemma, $\det A = \sum_j A_{ij} \det B_j = \sum_j A_{ij} (-1)^{i+j} \det(\tilde{B}_j)_{ij} = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$, as $(\tilde{B}_j)_{ij} = \tilde{A}_{ij}$. ■

Corollary

A matrix with two identical rows has determinant 0.

Proof. Induction on n; $n = 2$ is clear. For $n \geq 3$ and rows $r \neq s$ equal, expand along a row $i \notin \{r, s\}$: each $\tilde{A}_{ij}$ has two equal rows, so every term vanishes. ■

Theorem 4.5

Interchanging two rows multiplies the determinant by -1.

Proof. Let rows $r < s$ of A be u, v, interchanged in B. By Theorem 4.3 and the Corollary, $0 = \det(\dots, u+v, \dots, u+v, \dots) = \det(\dots, u, \dots, u, \dots) + \det(\dots, u, \dots, v, \dots) + \det(\dots, v, \dots, u, \dots) + \det(\dots, v, \dots, v, \dots) = 0 + \det A + \det B + 0$. ■

Theorem 4.6

Adding a multiple of one row to another leaves the determinant unchanged.

Proof. Let B be A with k·row r added to row s, and C be A with row s replaced by row r. Then $\det B = \det A + k \det C = \det A$ (Theorem 4.3; Corollary to Theorem 4.4). ■

Corollary

If $\operatorname{rank} A < n$, then $\det A = 0$.

Proof. The rows are dependent (Corollary 2 of Theorem 3.6): $a_r = \sum_{i \neq r} c_i a_i$. Subtracting each c_i·row i from row r gives a zero row with the same det (Theorem 4.6); apply the Corollary to Theorem 4.3. ■

Row operations: type 1 multiplies det by -1, type 2 by k, type 3 by 1. An upper triangular det is the diagonal product (expand along the last row repeatedly), so reduce to triangular form.

Example 5

For B of Example 2: swap rows $1, 2$, clear column 1, reach $\begin{pmatrix} -2 & -3 & -5 \\ 0 & 1 & 3 \\ 0 & 0 & 24 \end{pmatrix}$. So $\det B = -(-2)(1)(24) = 48$.

Example 6

For $\begin{pmatrix} 2 & 3 & 1 & 1 \\ -2 & -2 & 2 & -1 \\ 4 & 7 & 2 & 3 \\ -6 & -9 & 5 & 1 \end{pmatrix}$, type 3 operations give a triangular matrix with diagonal product $\det = -40$.

4.3 Properties of determinants

Motivation. det is multiplicative, detects invertibility, ignores transposition, gives Cramer's rule and measures volume.

By the row-operation rules, $\det E = -1, k, 1$ for elementary E of types $1, 2, 3$, and $\det(EA) = \det E \det A$.

Theorem 4.7

$\det(AB) = \det(A) \det(B)$ for $A, B \in \mathrm{M}_{n \times n}(\mathbb{F})$.

Proof.

Step 1 ($\operatorname{rank} A < n$). $\operatorname{rank}(AB) \leq \operatorname{rank} A < n$ (Theorem 3.7), so both sides are 0 (Corollary to Theorem 4.6).

Step 2 ($\operatorname{rank} A = n$). $A = E_m \cdots E_1$ (Corollary 3 of Theorem 3.6); by $\det(EM) = \det E \det M$, $\det(AB) = \det E_m \cdots \det E_1 \det B$ and $\det A = \det E_m \cdots \det E_1$. ■

Corollary

A is invertible iff $\det A \neq 0$, and then $\det(A^{-1}) = \frac{1}{\det A}$.

Proof. If A is not invertible, $\operatorname{rank} A < n$ and $\det A = 0$. If A is invertible, $1 = \det I = \det(AA^{-1}) = \det A \det A^{-1}$. ■

Theorem 4.8

$\det(A^t) = \det(A)$.

Proof. If A is not invertible, neither is A^t ($\operatorname{rank} A^t = \operatorname{rank} A$): both are 0. Otherwise $A = E_m \cdots E_1$, $A^t = E_1^t \cdots E_m^t$, and $\det E_i^t = \det E_i$ (types 1, 2 symmetric; type 3 transposes to type 3). By Theorem 4.7, $\det A^t = \prod \det E_i^t = \prod \det E_i = \det A$. ■

So all row statements hold for columns.

Theorem 4.9 (Cramer's rule)

If $Ax = b$ is an $n \times n$ system with $\det A \neq 0$, it has a unique solution, and

$$x_k = \frac{\det M_k}{\det A} \qquad (1 \leq k \leq n),$$

where M_k is A with column k replaced by b.

Proof. A is invertible, so the solution x is unique (Theorem 3.10). With $X_k = I_n$ with column k replaced by x, $AX_k = M_k$ (columns $Ae_j = a_j$, $j \neq k$, and $Ax = b$), and $\det X_k = x_k$ (expand along row k). So $\det M_k = \det A \cdot x_k$. ■

Example 1

$x_1 + 2x_2 + 3x_3 = 2$, $x_1 + x_3 = 3$, $x_1 + x_2 - x_3 = 1$: $\det A = 6$, $\det M_1 = 15$, $\det M_2 = -6$, $\det M_3 = 3$, so $(x_1, x_2, x_3) = (\frac{5}{2}, -1, \frac{1}{2})$.

Volume. For rows $a_1, \ldots, a_n \in \mathbb{R}^n$ of A, $|\det A|$ is the volume of the parallelepiped with sides $a_1, \ldots, a_n$ (as in Section 4.1.1, via Section 4.5).

Example 2

$a_1 = (1, -2, 1)$, $a_2 = (1, 0, -1)$, $a_3 = (1, 1, 1)$ give volume $\left|\det \begin{pmatrix} 1 & -2 & 1 \\ 1 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}\right| = 6 = \sqrt{6}\sqrt{2}\sqrt{3}$ (orthogonal sides).

An ordered basis β of $\mathbb{R}^n$ (rows of A) is **right-handed** if $\det A > 0$, **left-handed** if $\det A < 0$; coordinate changes with $\det Q > 0$ preserve this.

4.4 Summary: important facts about determinants

Motivation. The facts needed later, collected in one place.

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$:

1. $\det A = \sum_j (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any i, and $= \sum_i (-1)^{i+j} A_{ij} \det \tilde{A}_{ij}$ for any j (Theorems 4.4, 4.8).
2. Row (column) operations of types $1, 2, 3$ multiply det by -1, k, 1.
3. det of an upper or lower triangular matrix is the product of the diagonal entries; $\det I = 1$.
4. Two identical rows (columns), or a zero row, give det $= 0$.

5. $\det(AB) = \det A \det B$; $\det A^t = \det A$.

6. A is invertible iff $\det A \neq 0$, and then $\det A^{-1} = (\det A)^{-1}$.

7. $\det(Q^{-1}AQ) = \det Q^{-1} \det A \det Q = \det A$, so the **determinant of a linear operator** $\det \mathsf{T} = \det[\mathsf{T}]_\beta$ is independent of β.

8. For $A = \left(\begin{smallmatrix} B & C \\ O & D \end{smallmatrix}\right)$ with square B, D, $\det A = \det B \det D$.

Proof of 3 and 8 [Supplement].

3. Upper triangular: $\det A = A_{11} \det \tilde{A}_{11}$ (first column), $\tilde{A}_{11}$ upper triangular; induct. Lower: transpose.

8. Induction on the size k of B, expanding along column 1. For $k = 1$, $\det A = B_{11} \det D$. For $k > 1$, only rows $i \leq k$ contribute, and $\tilde{A}_{i1} = \left(\begin{smallmatrix} \tilde{B}_{i1} & C' \\ O & D \end{smallmatrix}\right)$, so $\det A = \sum_{i \leq k} (-1)^{i+1} B_{i1} \det \tilde{B}_{i1} \det D = \det B \det D$. ■

4.5 A characterization of the determinant *

Motivation. Row-linearity, vanishing on equal adjacent rows and $\det I = 1$ characterise det, explaining signed volume and multiplicativity.

Definition (n-linear; alternating)

$\delta : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is **n-linear** if linear in each row with the others fixed, and **alternating** if $\delta(A) = 0$ whenever two adjacent rows of A are identical.

Example 1

$\delta(A) = 0$ is n-linear.

Example 2

$\delta_j(A) = A_{1j}A_{2j}\cdots A_{nj}$ (product of column j) is n-linear (linear in A_{rj}).

Example 3

$\delta(A) = A_{11}A_{22}\cdots A_{nn}$ is n-linear.

Example 4

tr is not n-linear for $n \geq 2$: $\mathrm{tr}(2I) = 2\mathrm{tr}(I)$, not $2^n\mathrm{tr}(I)$.

det is n-linear (Theorem 4.3) and alternating (Corollary to Theorem 4.4).

Theorem 4.10

Let δ be alternating and n-linear.
(a) Interchanging two rows of A changes $\delta(A)$ to $-\delta(A)$.
(b) If A has two identical rows, $\delta(A) = 0$.

Proof.

(a) For adjacent rows u, v, Theorem 4.5's computation uses only adjacent equal rows: $0 = \delta(\dots, u+v, u+v, \dots) = \delta(A) + \delta(B)$. Swapping rows $r < s$ takes $2(s-r)-1$ adjacent swaps (row r down to s, then row s up to r), an odd number.

(b) Swaps make the identical rows adjacent, changing δ only by a sign; the result is 0. ■

Corollary 1

If B is A with a multiple of a row added to another row, then $\delta(B) = \delta(A)$.

Proof. As for Theorem 4.6, using Theorem 4.10(b). ■

Corollary 2

If $\operatorname{rank} M < n$, then $\delta(M) = 0$.

Proof. As for the Corollary to Theorem 4.6: type 3 operations (Corollary 1) produce a zero row, where $\delta = 0$ by linearity. ■

Corollary 3

For elementary E_1, E_2, E_3 of types 1, 2, 3, with E_2 multiplying a row of I by k: $\delta(E_1) = -\delta(I)$, $\delta(E_2) = k\delta(I)$, $\delta(E_3) = \delta(I)$.

Proof. Theorem 4.10(a), linearity in the scaled row, and Corollary 1, applied to I. ■

Theorem 4.11

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(AB) = \delta(A)\delta(B)$.

Proof [Supplement]. By Theorem 4.10, linearity and Corollary 1, $\delta(EB) = \delta(E)\delta(B)$ for elementary E ($\delta(E) = -1, k, 1$ by Corollary 3). Repeat the proof of Theorem 4.7, using Corollary 2 if $\operatorname{rank} A < n$. ■

Theorem 4.12

If δ is alternating, n-linear and $\delta(I) = 1$, then $\delta(A) = \det(A)$ for all A.

Proof. If $\operatorname{rank} A < n$, both vanish (Corollary 2 and the Corollary to Theorem 4.6). Otherwise $A = E_m \cdots E_1$, and by Theorem 4.11 and Corollary 3, $\delta(A) = \prod \delta(E_i) = \prod \det(E_i) = \det A$. ■